

Lipids Analysis

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In this document, I'm going to analyse the relationship between the lipids used for formulation of nanoparticles.

PCA of the lipid pairs

First, I'm going to make a principal component analysis of the models coefficients and see what parameter each lipid pair approximates.

```
library(magick)
```

Warning: pacote 'magick' foi compilado no R versão 4.5.2

Linking to ImageMagick 6.9.13.29

Enabled features: cairo, freetype, fftw, ghostscript, heic, lcms, pango, raw, rsvg, webp

Disabled features: fontconfig, x11

```
coefData <- read.csv("C:/Users/luizr/Documents/Documents/Documents - Cópia/NanoIA/Topicos_01/coefData.csv")
```

```
head(coefData)
```

	Combination	Model	Bacuri	Açaí	Pracaxi	Cacau	Tucumã	Cupuaçu	Pequi	Manga	
1	Bacuri-Açaí	Linear	1	1	0	0	0	0	0	0	
2	Bacuri-Pracaxi	Linear	1	0	1	0	0	0	0	0	
3	Cacau-Tucumã	Linear	0	0	0	1	1	0	0	0	
4	Cupuaçu-Açaí	Linear	0	1	0	0	0	1	0	0	
5	Cupuaçu-Cacau	Linear	0	0	0	1	0	1	0	0	
6	Cupuaçu-Pequi	Quadratic	0	0	0	0	0	1	1	0	
	Andiroba	Murumuru	Buriti	Jambu	Urucum	Ucuuba	b1	b2	b3	b12	b13
1	0	0	0	0	0	0	2.3411	-3.464	-1.527	0.00	0.00

2	0	0	0	0	0	0	0.8600	0.410	0.010	0.00	0.00
3	0	0	0	0	0	0	0.4300	0.270	0.140	0.00	0.00
4	0	0	0	0	0	0	0.7000	1.160	-0.260	0.00	0.00
5	0	0	0	0	0	0	1.2200	-0.280	0.190	0.00	0.00
6	0	0	0	0	0	0	-0.5100	0.830	-1.840	-3.92	7.41

```

      b23    R.
1 0.00 0.78
2 0.00 0.33
3 0.00 0.06
4 0.00 0.36
5 0.00 0.75
6 5.15 0.58

```

```

coefDataPca <- coefData[, c("b1", "b2", "b3", "b12", "b13", "b23")]

rownames(coefDataPca) <- coefData$Combination

coefPca <- prcomp(
  x = coefDataPca,
  retx = TRUE,
  center = TRUE,
  scale. = TRUE
)

coefPca

```

Standard deviations (1, ..., p=6):

```
[1] 1.8033931 1.1619053 1.0462492 0.4952753 0.2141901 0.1092559
```

Rotation (n x k) = (6 x 6):

	PC1	PC2	PC3	PC4	PC5	PC6
b1	-0.4178847	0.55475882	-0.03028403	-0.06722947	-0.4964542	0.51547200
b2	-0.1800548	-0.60155349	0.59884031	0.11071795	-0.4756009	0.09300166
b3	-0.5091257	-0.25335513	-0.11128759	0.40593384	0.5534962	0.43940532
b12	0.1823600	-0.44283703	-0.75146256	0.06865526	-0.3990617	0.20489122
b13	0.5252229	-0.08339741	0.21279849	-0.38291162	0.2204509	0.69042211
b23	0.4739510	0.25125178	0.13456987	0.81676471	-0.1145871	0.11789474

```
summary(coefPca)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6
Standard deviation	1.803	1.162	1.0462	0.49528	0.21419	0.10926
Proportion of Variance	0.542	0.225	0.1824	0.04088	0.00765	0.00199
Cumulative Proportion	0.542	0.767	0.9495	0.99036	0.99801	1.00000

Now, lets try to do some plots,

starting with the scree plot of the explained variance per dimension.

```
#| warning: False
#| message: False

library(factoextra)
```

Warning: pacote 'factoextra' foi compilado no R versão 4.5.3

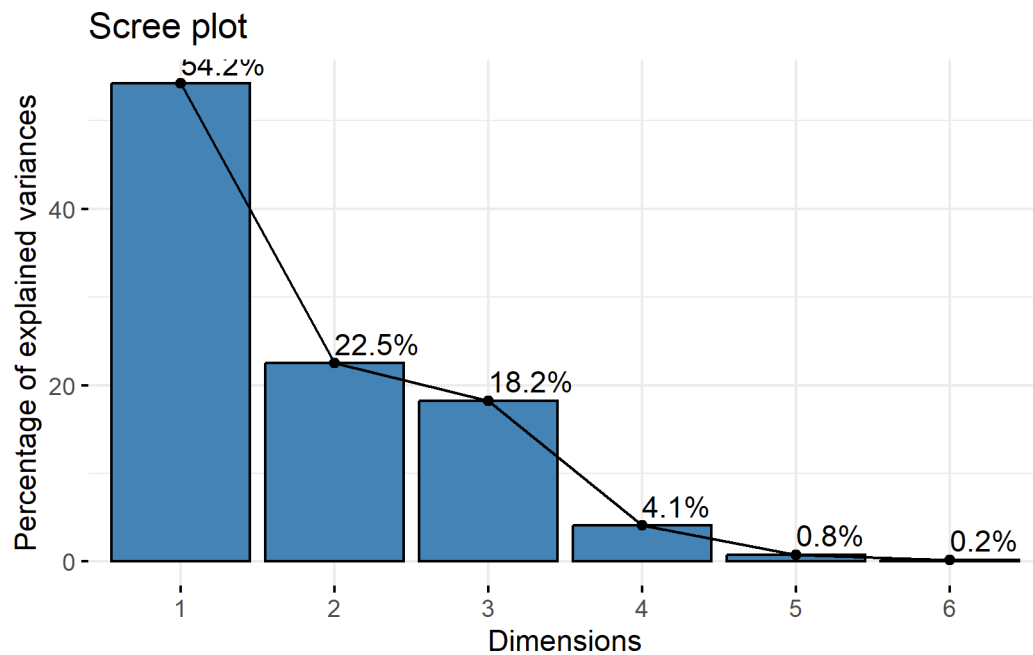
Carregando pacotes exigidos: ggplot2

Welcome to factoextra!

Want to learn more? See two factoextra-related books at <https://www.datanovia.com/en/product>.

```
screePlot <- fviz_eig(coefPca,
  # choice = "eigenvalue",
  barcolor = "black",
  addlabels = TRUE,
  ggtheme = theme_minimal(),
  # parallel = TRUE
) #+ theme_void()

screePlot
```



```
ggsave(
  "pcaCoefVar.png",
  plot = screePlot,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 600
)
```

First, I'll plot the parameter's vectors

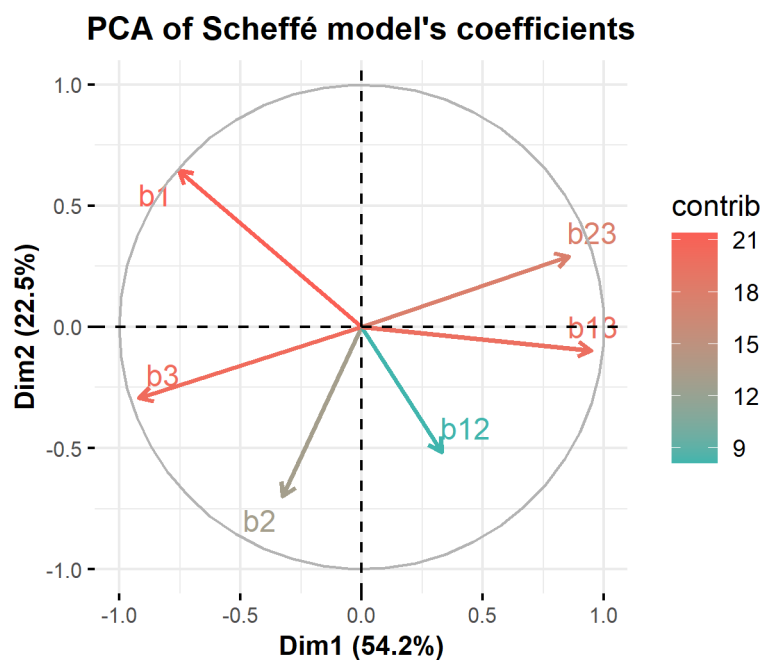
```
PcaVar <- fviz_pca_var(
  X = coefPca,
  repel = TRUE,
  arrowsize = 0.8,
  col.var = "contrib",
  gradient.cols = c("#40B5AD", "#FA5F55")
) +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 12
    )
  )
```

```

),
axis.title = element_text(
  face = "bold",
  size = 10
),
axis.text = element_text(size = 8)
) +
labs(
  title = "PCA of Scheffé model's coefficients"
)

```

PcaVar



```

ggsave(
  "pcaVar.png",
  plot = PcaVar,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 600
)

```

Now, I'll plot just the lipid pairs

```

cluster <- c(
  "Cupuaçu-Cacau",
  "Bacuri-Pracaxi",
  "Cacau-Tucumã",
  "Cupuaçu-Tucumã",
  "Cupuaçu-Açaí",
  "Tucumã-Urucum",
  "Manga-Andiroba",
  "Murumuru-Jambu",
  "Manga-Pracaxi",
  "Murumuru-Cupuaçu",
  "Ucuuba-Andiroba"
)

coord <- as.data.frame(coefPca$x)

head(coefPca)

```

\$sdev

```
[1] 1.8033931 1.1619053 1.0462492 0.4952753 0.2141901 0.1092559
```

\$rotation

	PC1	PC2	PC3	PC4	PC5	PC6
b1	-0.4178847	0.55475882	-0.03028403	-0.06722947	-0.4964542	0.51547200
b2	-0.1800548	-0.60155349	0.59884031	0.11071795	-0.4756009	0.09300166
b3	-0.5091257	-0.25335513	-0.11128759	0.40593384	0.5534962	0.43940532
b12	0.1823600	-0.44283703	-0.75146256	0.06865526	-0.3990617	0.20489122
b13	0.5252229	-0.08339741	0.21279849	-0.38291162	0.2204509	0.69042211
b23	0.4739510	0.25125178	0.13456987	0.81676471	-0.1145871	0.11789474

\$center

	b1	b2	b3	b12	b13	b23
	0.9718294	0.4891765	0.3448824	-0.7388235	-1.3400000	-0.9623529

\$scale

	b1	b2	b3	b12	b13	b23
	1.206349	1.214323	1.166249	2.086872	4.071098	3.261126

\$x

	PC1	PC2	PC3	PC4	PC5
Bacuri-Açaí	1.3063094	2.88457250	-1.96153988	-0.948991617	-0.006125201
Bacuri-Pracaxi	0.5739716	-0.04953989	-0.16057200	0.021748383	-0.184436261

Cacau-Tucumã	0.6869326	-0.20616989	-0.23122301	0.078196246	0.109053161
Cupuaçu-Açaí	0.6360579	-0.43599999	0.23906942	0.005069283	-0.540475823
Cupuaçu-Cacau	0.4729973	0.41872202	-0.52705724	0.001426031	0.023083315
Cupuaçu-Pequi	3.1557911	0.59108902	2.26886850	-0.043609636	0.306765205
Cupuaçu-Tucumã	0.7176457	-0.20775296	-0.18670819	0.045176546	0.021200860
Manga-Andiroba	0.6040080	-0.64791591	0.04858257	0.208017824	0.062169661
Manga-Pracaxi	0.5965942	-0.67268498	0.07323993	0.212576656	0.042586705
Murumuru-Açaí	-3.3429548	-1.69819424	0.69489477	-1.369045919	0.166402844
Murumuru-Buriti	-2.5543077	0.01977843	-1.62113813	0.928558475	0.257567851
Murumuru-Cupuaçu	0.7226342	-0.52353248	-0.03211754	0.128802156	0.067420374
Murumuru-Jambu	0.7139142	-0.48570934	-0.10335849	0.145909219	0.164534365
Tucumã-Tucumã	-2.1109671	-0.59980085	-0.72730293	0.069433063	-0.114127439
Tucumã-Urucum	0.7261887	-0.30255753	-0.17039845	0.080433227	0.092255625
Ucuuba-Andiroba	0.7578602	-0.76318214	0.32973098	0.069318232	-0.372814992
Ucuuba-Buriti	-3.6626755	2.67887825	2.06702967	0.366981830	-0.095060250

PC6

Bacuri-Açaí	-0.088359037
Bacuri-Pracaxi	0.154559619
Cacau-Tucumã	0.009078632
Cupuaçu-Açaí	0.041904803
Cupuaçu-Cacau	0.323360368
Cupuaçu-Pequi	-0.037713206
Cupuaçu-Tucumã	-0.018458766
Manga-Andiroba	0.019531552
Manga-Pracaxi	0.023360913
Murumuru-Açaí	0.037689711
Murumuru-Buriti	-0.003844742
Murumuru-Cupuaçu	-0.076423115
Murumuru-Jambu	-0.064783992
Tucumã-Tucumã	-0.134894907
Tucumã-Urucum	-0.044570399
Ucuuba-Andiroba	-0.136118847
Ucuuba-Buriti	-0.004318584

```
coord$lipid <- rownames(coord)

coord$group <- coord$lipid %in% cluster

groupCoord <- subset(coord, group)

hull <- groupCoord[
  chull(
```

```

      groupCoord$PC1,
      groupCoord$PC2
    ),
  ]

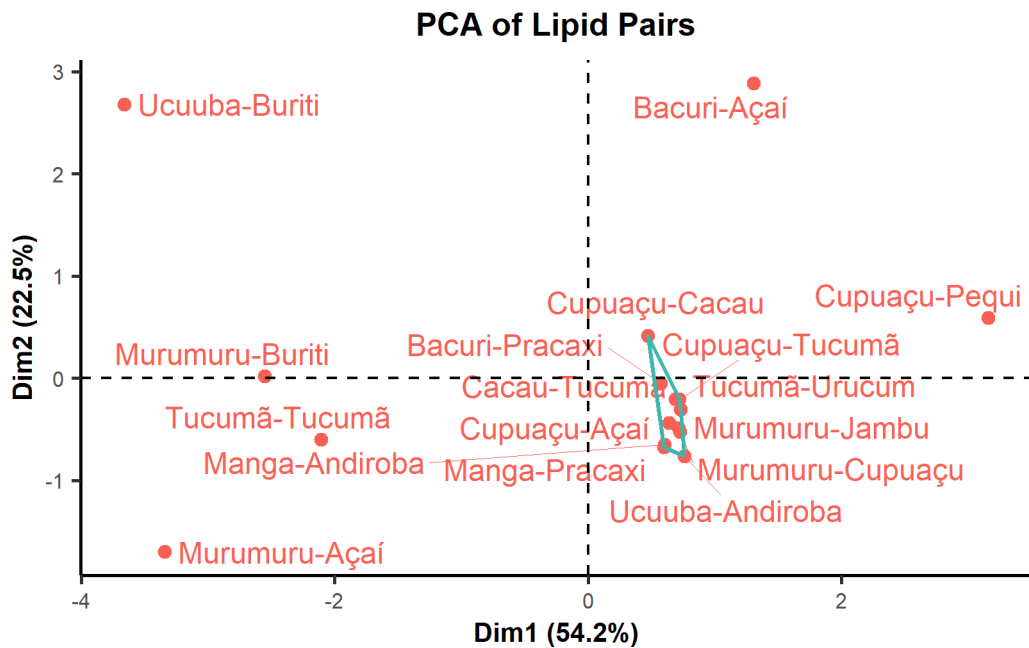
pcaInd <- fviz_pca_ind(
  X = coefPca,
  repel = TRUE,
  col.ind = "#FA5F55",
  #fill.ind = "contrib",
  pointsize = 2,
  arrowsize = 0.8
) + theme_classic(base_size = 14) +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 12
    ),
    axis.title = element_text(
      face = "bold",
      size = 10
    ),
    axis.text = element_text(size = 8)
  ) +
  labs(
    title = "PCA of Lipid Pairs"
  ) + geom_polygon(
    data = hull,
    aes(
      x = PC1,
      y = PC2
    ),
    inherit.aes = FALSE,
    fill = NA,
    color = "#40B5AD",
    linewidth = 0.7
  )
# stat_ellipse(
#   data = subset(coord, group),

```



```
# aes(
#   x = PC1,
#   y = PC2
# ),
# linewidth = 1
# )
```

pcaInd



```
ggsave(
  "pcaInd.png",
  plot = pcaInd,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 600
)
```

Now I shall plot the bidimensional graph with the parameter's vectors and lipid pair points.

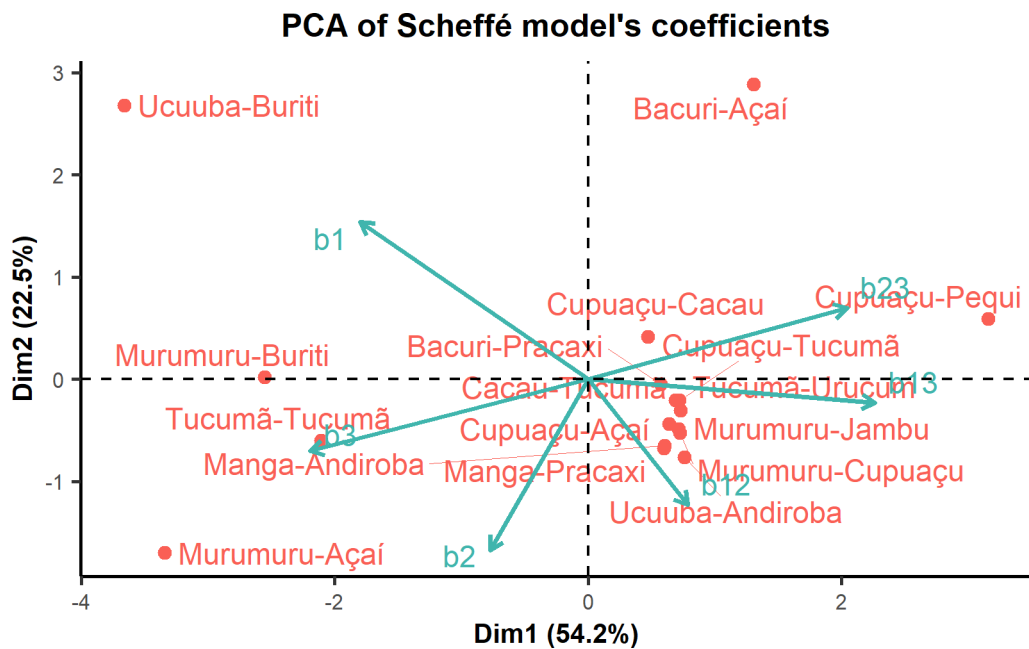
```
pcaPlot <- fviz_pca(
  X = coefPca,
  repel = TRUE,
```

```

col.var = "#40B5AD",
col.ind = "#FA5F55",
#fill.ind = "contrib",
pointsize = 2,
arrowsize = 0.8
) + theme_classic(base_size = 14) +
theme(
  plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 12
  ),
  axis.title = element_text(
    face = "bold",
    size = 10
  ),
  axis.text = element_text(size = 8)
) +
labs(
  title = "PCA of Scheffé model's coefficients"
)

```

pcaPlot



```

ggsave(
  "pcaLipidPair.png",
  plot = pcaPlot,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 600
)

```

Now I'll plot the lipid pairs with their **Scheffé equation's degree fitted**.

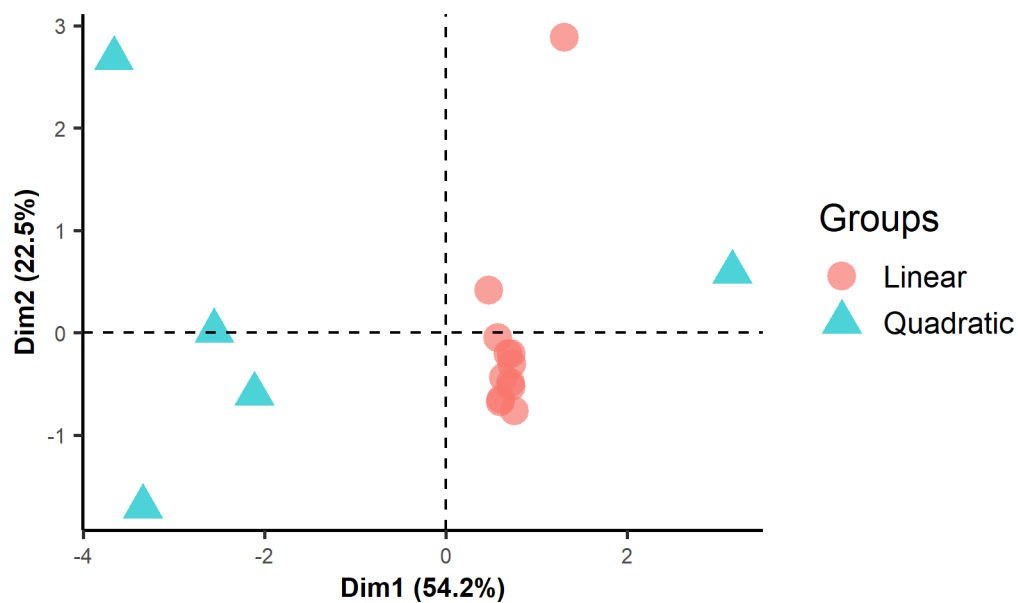
```

pcaModel <- fviz_pca_ind(
  X = coefPca,
  geom = "point",
  repel = TRUE,
  col.ind = "#FA5F55",
  habillage = coefData$Model,
  mean.point = FALSE,
  alpha.ind = 0.7,
  #addEllipses = TRUE,
  #fill.ind = "contrib",
  pointsize = 5,
  arrowsize = 0.8
) + theme_classic(base_size = 14) +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 12
    ),
    axis.title = element_text(
      face = "bold",
      size = 10
    ),
    axis.text = element_text(size = 8)
  ) +
  labs(
    title = "PCA of Lipid Pairs by Degree of Scheffé's Equation"
  )

pcaModel

```

PCA of Lipid Pairs by Degree of Scheffé's Equation



```
ggsave(
  "pcaModel.png",
  plot = pcaModel,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 600
)
```

Now, I'll plot the lipid pairs with their nature as butter or oil, to identify some relationship.

```
coefDataComp <- read.csv("C:/Users/luisr/Documents/Documents/Documentos - Copia/NanoIA/Topi
head(coefDataComp)
```

	Combination	Model	Type	FP	deltaFP	esfa	deltaSfa	eufa
1	Bacuri-Açaí	Linear	butter-oil	35-4	31.0	75-28	47	25-72
2	Bacuri-Pracaxi	Linear	butter-oil	35-18.5	16.5	75-35	40	25-65
3	Cacau-Tucumã	Linear	butter-oil	35-30	5.0	73-27	46	27-73
4	Cupuaçu-Açaí	Linear	butter-oil	27-4	23.0	55-28	27	45-72
5	Cupuaçu-Cacau	Linear	butter-butter	27-35	-8.0	55-35	20	45-65
6	Cupuaçu-Pequi	Quadratic	butter-oil	27-21	6.0	55-42	13	45-58
	FA Bacuri Açaí Pracaxi Cacau Tucumã Cupuaçu Pequi Manga Andiroba							

1	C16-C18:1	1	1	0	0	0	0	0	0	0	0
2	C16-C18:1	1	0	1	0	0	0	0	0	0	0
3	C16-C18:1	0	0	0	1	1	0	0	0	0	0
4	C18:1-C18:1	0	1	0	0	0	1	0	0	0	0
5	C16-C18:1	0	0	0	1	0	1	0	0	0	0
6	C18:1-C18:1	0	0	0	0	0	1	1	0	0	0
	Murumuru	Buriti	Jambu	Urucum	Ucuuba	b1	b2	b3	b12	b13	b23
1	0	0	0	0	0	2.3411	-3.464	-1.527	0.00	0.00	0.00
2	0	0	0	0	0	0.8600	0.410	0.010	0.00	0.00	0.00
3	0	0	0	0	0	0.4300	0.270	0.140	0.00	0.00	0.00
4	0	0	0	0	0	0.7000	1.160	-0.260	0.00	0.00	0.00
5	0	0	0	0	0	1.2200	-0.280	0.190	0.00	0.00	0.00
6	0	0	0	0	0	-0.5100	0.830	-1.840	-3.92	7.41	5.15

R.

1	0.78
2	0.33
3	0.06
4	0.36
5	0.75
6	0.58

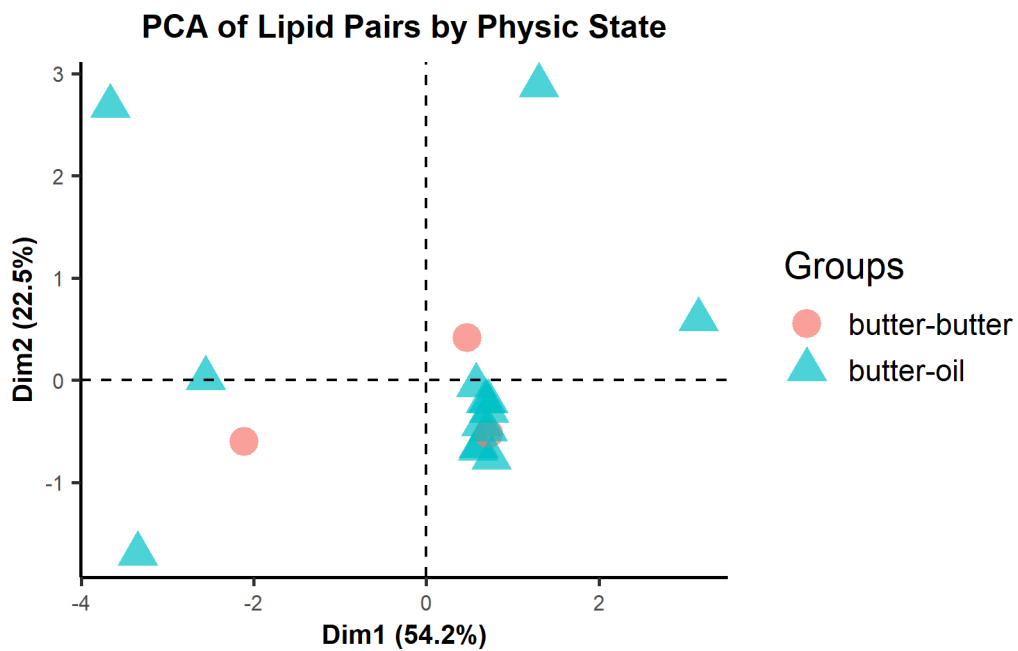
```
pcaType <- fviz_pca_ind(
  X = coefPca,
  geom = "point",
  repel = TRUE,
  col.ind = "#FA5F55",
  habillage = coefDataComp$Type,
  mean.point = FALSE,
  alpha.ind = 0.7,
  #addEllipses = TRUE,
  #fill.ind = "contrib",
  pointsize = 5,
  arrowsize = 0.8
) + theme_classic(base_size = 14) +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 12
    ),
    axis.title = element_text(
      face = "bold",
```

```

    size = 10
  ),
  axis.text = element_text(size = 8)
) +
labs(
  title = "PCA of Lipid Pairs by Physic State"
)

```

pcaType



```

ggsave(
  "pcaType.png",
  plot = pcaType,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 600
)

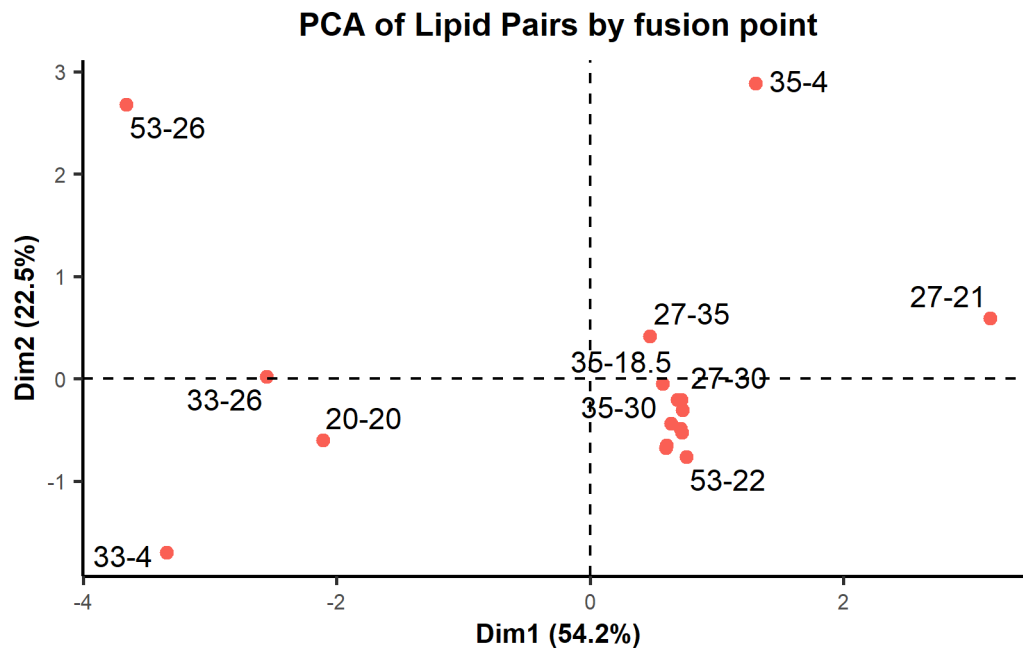
```

Now I'll plot with the fusion point:

```
#install.packages("ggrepel")  
library(ggrepel)
```

Warning: pacote 'ggrepel' foi compilado no R versão 4.5.3

```
pcaFP <- fviz_pca_ind(  
  X = coefPca,  
  repel = TRUE,  
  col.ind = "#FA5F55",  
  label = "none",  
  #fill.ind = "contrib",  
  pointsize = 2,  
  arrowsize = 0.8  
) + theme_classic(base_size = 14) +  
  theme(  
    plot.title = element_text(  
      hjust = 0.5,  
      face = "bold",  
      size = 12  
    ),  
    axis.title = element_text(  
      face = "bold",  
      size = 10  
    ),  
    axis.text = element_text(size = 8)  
  ) +  
  labs(  
    title = "PCA of Lipid Pairs by fusion point"  
  ) +  
  geom_text_repel(  
    aes(  
      x = coefPca$x[,1],  
      y = coefPca$x[,2],  
      label = coefDataComp$FP  
    )  
  )  
pcaFP
```



```
ggsave(
  "pcaFP.png",
  plot = pcaFP,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 600
)
```

now, I'll plot the **difference of fusion points**.

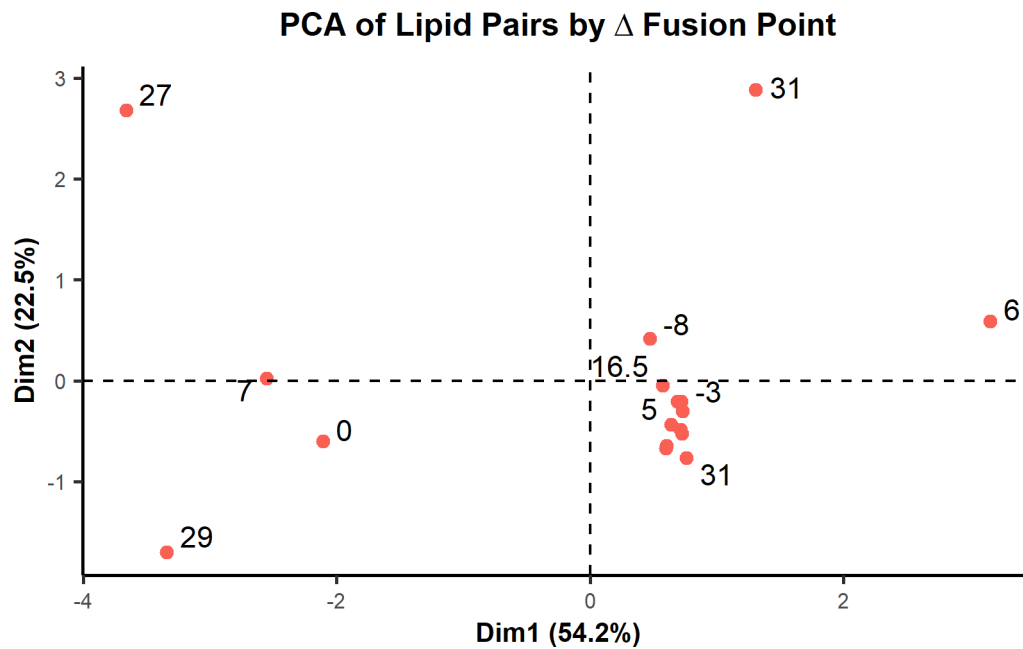
```
pcaDeltaFP <- fviz_pca_ind(
  X = coefPca,
  repel = TRUE,
  col.ind = "#FA5F55",
  label = "none",
  #fill.ind = "contrib",
  pointsize = 2,
  arrowsize = 0.8
) + theme_classic(base_size = 14) +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
```



```

    size = 12
  ),
  axis.title = element_text(
    face = "bold",
    size = 10
  ),
  axis.text = element_text(size = 8)
) +
labs(
  title = expression(
    bold(
      "PCA of Lipid Pairs by"~Delta~"Fusion Point"
    )
  )
)+
geom_text_repel(
  aes(
    x = coefPca$x[,1],
    y = coefPca$x[,2],
    label = coefDataComp$deltaFP
  )
)
pcaDeltaFP

```



```
ggsave(
  "pcaDeltaFP.png",
  plot = pcaDeltaFP,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 600
)
```

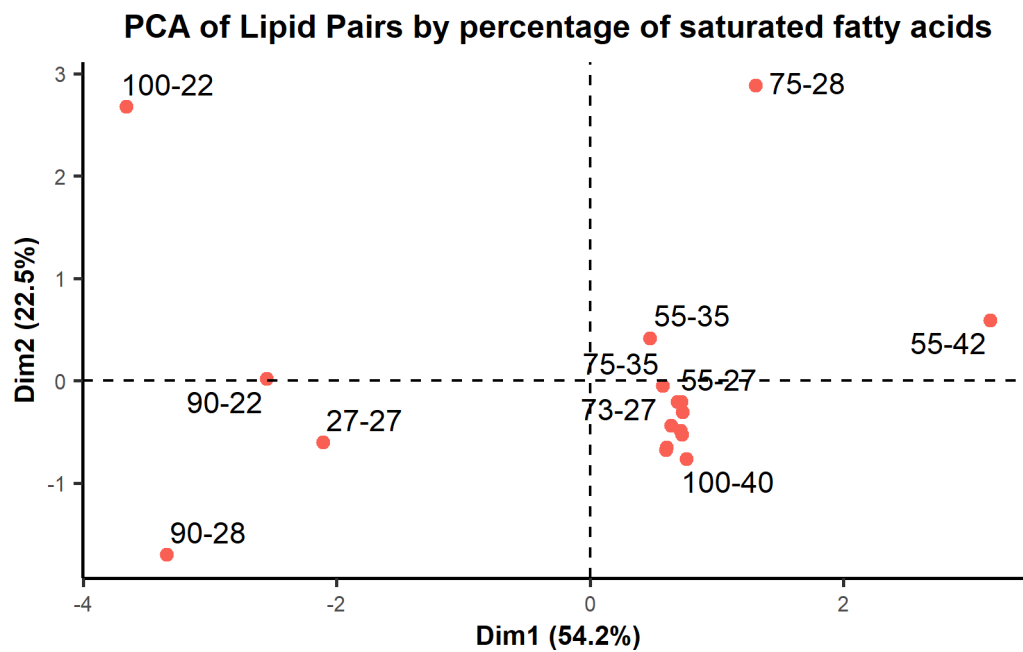
Lets now plot the points with the **percentage of saturated fatty acids**:

```
pcaSfa <- fviz_pca_ind(
  X = coefPca,
  repel = TRUE,
  col.ind = "#FA5F55",
  label = "none",
  #fill.ind = "contrib",
  pointsize = 2,
  arrowsize = 0.8
) + theme_classic(base_size = 14) +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
```

```

    size = 12
  ),
  axis.title = element_text(
    face = "bold",
    size = 10
  ),
  axis.text = element_text(size = 8)
) +
labs(
  title = "PCA of Lipid Pairs by percentage of saturated fatty acids"
)+
geom_text_repel(
  aes(
    x = coefPca$x[,1],
    y = coefPca$x[,2],
    label = coefDataComp$esfa
  )
)
pcaSfa

```



```

ggsave(
  "pcaSfa.png",

```

```

plot = pcaSfa,
width = 18,
height = 14,
units = "cm",
dpi = 600
)

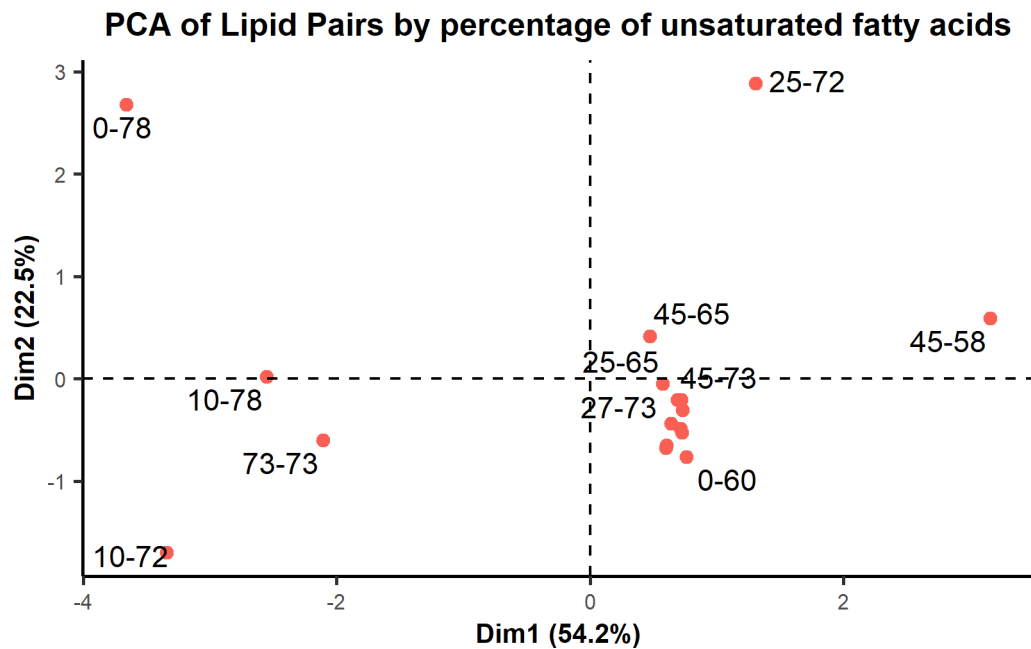
```

Now, I'll plot with the percentage of **unsaturated fatty acids**.

```

pcaUfa <- fviz_pca_ind(
  X = coefPca,
  repel = TRUE,
  col.ind = "#FA5F55",
  label = "none",
  #fill.ind = "contrib",
  pointsize = 2,
  arrowsize = 0.8
) + theme_classic(base_size = 14) +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 12
    ),
    axis.title = element_text(
      face = "bold",
      size = 10
    ),
    axis.text = element_text(size = 8)
  ) +
  labs(
    title = "PCA of Lipid Pairs by percentage of unsaturated fatty acids"
  ) +
  geom_text_repel(
    aes(
      x = coefPca$x[,1],
      y = coefPca$x[,2],
      label = coefDataComp$eufa
    )
  )
pcaUfa

```



```
ggsave(
  "pcaUfa.png",
  plot = pcaUfa,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 600
)
```

Plotting now the **difference of the percentage of saturated fatty acids**.

```
clusterSfa <- c(
  "Ucuuba-Buriti",
  "Murumuru-Buriti",
  "Murumuru-Açaí"
)

coordSfa <- as.data.frame(coefPca$x)

head(coefPca)
```

```
$sdev
[1] 1.8033931 1.1619053 1.0462492 0.4952753 0.2141901 0.1092559
```

\$rotation

	PC1	PC2	PC3	PC4	PC5	PC6
b1	-0.4178847	0.55475882	-0.03028403	-0.06722947	-0.4964542	0.51547200
b2	-0.1800548	-0.60155349	0.59884031	0.11071795	-0.4756009	0.09300166
b3	-0.5091257	-0.25335513	-0.11128759	0.40593384	0.5534962	0.43940532
b12	0.1823600	-0.44283703	-0.75146256	0.06865526	-0.3990617	0.20489122
b13	0.5252229	-0.08339741	0.21279849	-0.38291162	0.2204509	0.69042211
b23	0.4739510	0.25125178	0.13456987	0.81676471	-0.1145871	0.11789474

\$center

	b1	b2	b3	b12	b13	b23
	0.9718294	0.4891765	0.3448824	-0.7388235	-1.3400000	-0.9623529

\$scale

	b1	b2	b3	b12	b13	b23
	1.206349	1.214323	1.166249	2.086872	4.071098	3.261126

\$x

	PC1	PC2	PC3	PC4	PC5
Bacuri-Açaí	1.3063094	2.88457250	-1.96153988	-0.948991617	-0.006125201
Bacuri-Pracaxi	0.5739716	-0.04953989	-0.16057200	0.021748383	-0.184436261
Cacau-Tucumã	0.6869326	-0.20616989	-0.23122301	0.078196246	0.109053161
Cupuaçu-Açaí	0.6360579	-0.43599999	0.23906942	0.005069283	-0.540475823
Cupuaçu-Cacau	0.4729973	0.41872202	-0.52705724	0.001426031	0.023083315
Cupuaçu-Pequi	3.1557911	0.59108902	2.26886850	-0.043609636	0.306765205
Cupuaçu-Tucumã	0.7176457	-0.20775296	-0.18670819	0.045176546	0.021200860
Manga-Andiroba	0.6040080	-0.64791591	0.04858257	0.208017824	0.062169661
Manga-Pracaxi	0.5965942	-0.67268498	0.07323993	0.212576656	0.042586705
Murumuru-Açaí	-3.3429548	-1.69819424	0.69489477	-1.369045919	0.166402844
Murumuru-Buriti	-2.5543077	0.01977843	-1.62113813	0.928558475	0.257567851
Murumuru-Cupuaçu	0.7226342	-0.52353248	-0.03211754	0.128802156	0.067420374
Murumuru-Jambu	0.7139142	-0.48570934	-0.10335849	0.145909219	0.164534365
Tucumã-Tucumã	-2.1109671	-0.59980085	-0.72730293	0.069433063	-0.114127439
Tucumã-Urucum	0.7261887	-0.30255753	-0.17039845	0.080433227	0.092255625
Ucuuba-Andiroba	0.7578602	-0.76318214	0.32973098	0.069318232	-0.372814992
Ucuuba-Buriti	-3.6626755	2.67887825	2.06702967	0.366981830	-0.095060250
	PC6				
Bacuri-Açaí	-0.088359037				
Bacuri-Pracaxi	0.154559619				
Cacau-Tucumã	0.009078632				
Cupuaçu-Açaí	0.041904803				
Cupuaçu-Cacau	0.323360368				

Cupuaçu-Pequi	-0.037713206
Cupuaçu-Tucumã	-0.018458766
Manga-Andiroba	0.019531552
Manga-Pracaxi	0.023360913
Murumuru-Açaí	0.037689711
Murumuru-Buriti	-0.003844742
Murumuru-Cupuaçu	-0.076423115
Murumuru-Jambu	-0.064783992
Tucumã-Tucumã	-0.134894907
Tucumã-Urucum	-0.044570399
Ucuuba-Andiroba	-0.136118847
Ucuuba-Buriti	-0.004318584

```

coordSfa$lipid <- rownames(coordSfa)

coordSfa$group <- coordSfa$lipid %in% clusterSfa

groupCoordSfa <- subset(coordSfa, group)

hullSfa <- groupCoordSfa[
  chull(
    groupCoordSfa$PC1,
    groupCoordSfa$PC2
  ),
]

pcaDeltaSfa <- fviz_pca_ind(
  X = coefPca,
  repel = TRUE,
  col.ind = "#FA5F55",
  label = "none",
  #fill.ind = "contrib",
  pointsize = 2,
  arrowsize = 0.8
) + theme_classic(base_size = 14) +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 12
    )
  )

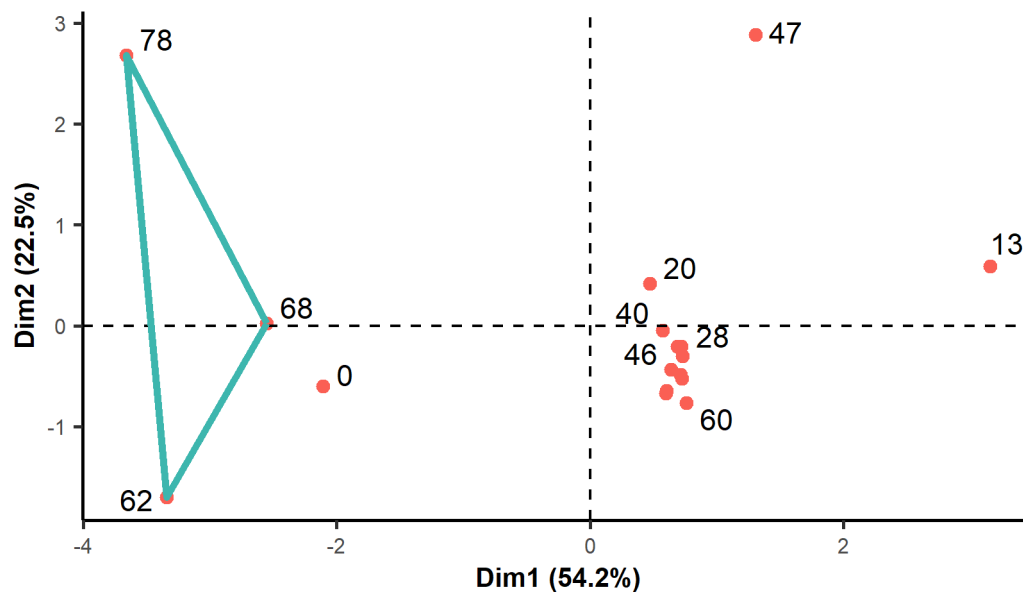
```

```

    ),
    axis.title = element_text(
      face = "bold",
      size = 10
    ),
    axis.text = element_text(size = 8)
  ) +
  labs(
    title = expression(
      bold(
        "PCA of Lipid Pairs by"~Delta~"Percentage of Saturated Fatty Acids"
      )
    )
  ) +
  geom_text_repel(
    aes(
      x = coefPca$x[,1],
      y = coefPca$x[,2],
      label = coefDataComp$deltaSfa
    )
  ) + geom_polygon(
    data = hullSfa,
    aes(
      x = PC1,
      y = PC2
    ),
    inherit.aes = FALSE,
    fill = NA,
    color = "#40B5AD",
    linewidth = 1.3
  )
pcaDeltaSfa

```


PCA of Lipid Pairs by Δ Percentage of Saturated Fatty Acids



```
ggsave(
  "pcaDeltaSfa.png",
  plot = pcaDeltaSfa,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 600
)
```

Now, I'll plot the points with the label of **major fatty acid per lipid**:

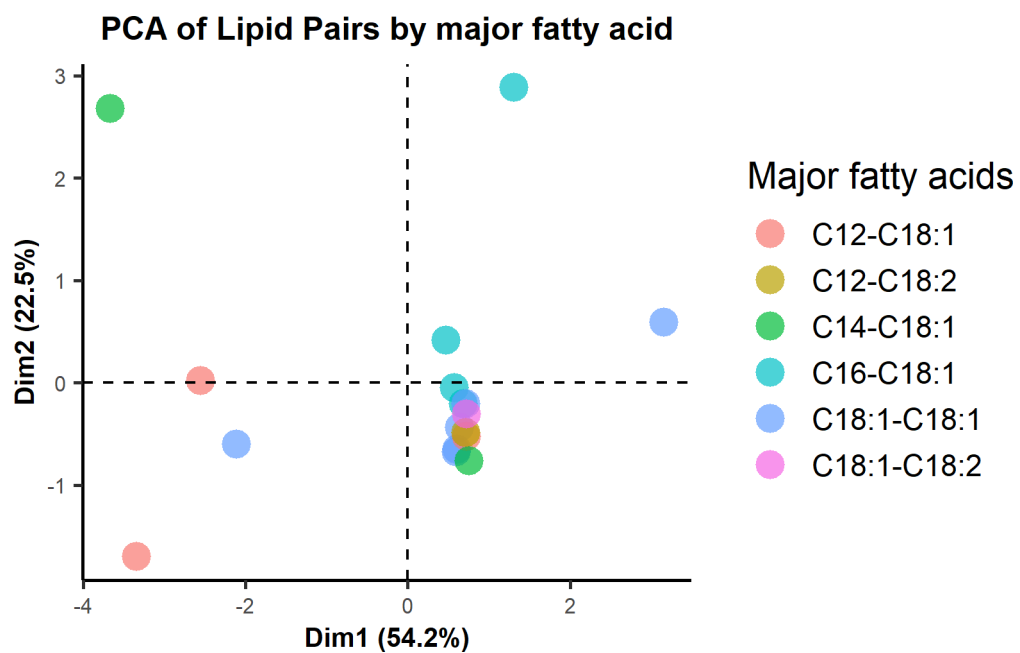
```
pcaFA <- fviz_pca_ind(
  X = coefPca,
  geom = "point",
  pointshape = 16,
  repel = TRUE,
  col.ind = "#FA5F55",
  habillage = coefDataComp$FA,
  mean.point = FALSE,
  alpha.ind = 0.7,
  #addEllipses = TRUE,
  #fill.ind = "contrib",
  pointsize = 5,
  arrowsize = 0.8
)
```

```

) + theme_classic(base_size = 14) +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 12
    ),
    axis.title = element_text(
      face = "bold",
      size = 10
    ),
    axis.text = element_text(size = 8)
  ) +
  labs(
    title = "PCA of Lipid Pairs by major fatty acid",
    color = "Major fatty acids"
  )

```

pcaFA



```

ggsave(
  "pcaFA.png",
  plot = pcaFA,

```

```
width = 18,
height = 14,
units = "cm",
dpi = 600
)
```

Now, I'll try to use *clustering* methods to separate the lipid pairs into **clusters** and try to extract some data.

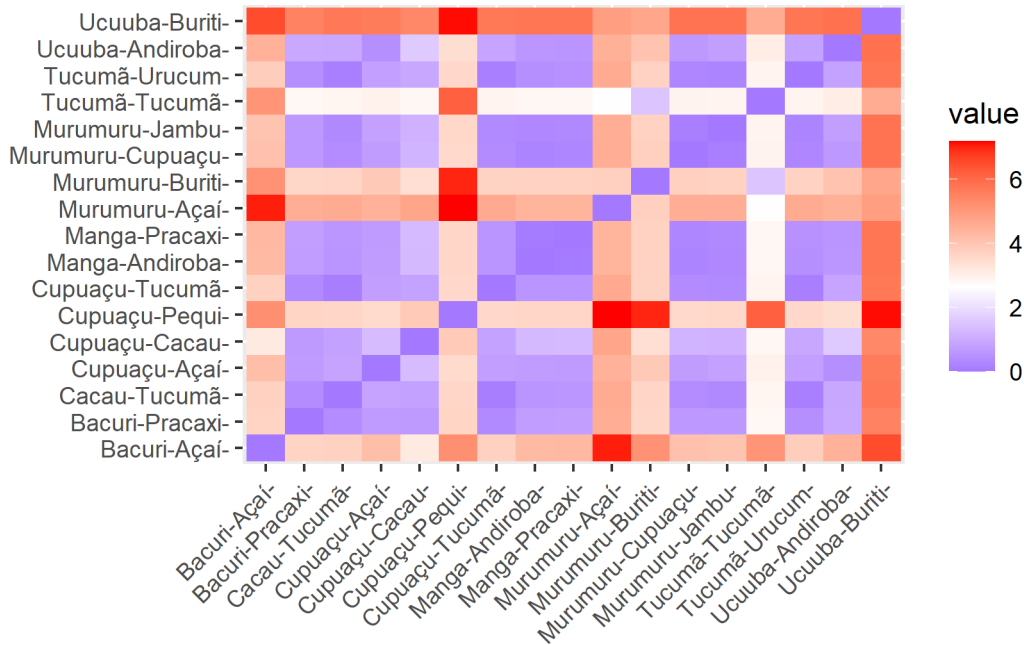
```
#| message: FALSE

distPair <- dist(
  scale(coefDataPca)
)
distPair
```

	Bacuri-Açaí	Bacuri-Pracaxi	Cacau-Tucumã	Cupuaçu-Açaí
Bacuri-Pracaxi	3.66359931			
Cacau-Tucumã	3.74275134	0.39086038		
Cupuaçu-Açaí	4.18698597	0.67279481	0.83958210	
Cupuaçu-Cacau	3.14741603	0.66011174	0.79739173	1.31943849
Cupuaçu-Pequi	5.24377592	3.64155155	3.61074197	3.50008249
Cupuaçu-Tucumã	3.74873742	0.34514865	0.11177658	0.74884772
Manga-Andiroba	4.28570522	0.71864110	0.54723684	0.69791908
Manga-Pracaxi	4.31997689	0.74071422	0.58410180	0.68442800
Murumuru-Açaí	7.06372093	4.56763093	4.62843828	4.47452129
Murumuru-Buriti	5.17968667	3.60096572	3.63784211	3.91641901
Murumuru-Cupuaçu	4.10435350	0.62571192	0.39146061	0.69824700
Murumuru-Jambu	4.04858717	0.63119659	0.32914821	0.80873188
Tucumã-Tucumã	5.13741139	2.81493654	2.88095436	2.95364624
Tucumã-Urucum	3.84367495	0.45490617	0.13302868	0.77916697
Ucuuba-Andiroba	4.47554800	0.95170410	0.93985147	0.44049288
Ucuuba-Buriti	6.53523160	5.52341107	5.71400017	5.64398141
	Cupuaçu-Cacau	Cupuaçu-Pequi	Cupuaçu-Tucumã	Manga-Andiroba
Bacuri-Pracaxi				
Cacau-Tucumã				
Cupuaçu-Açaí				
Cupuaçu-Cacau				
Cupuaçu-Pequi	3.90604288			
Cupuaçu-Tucumã	0.82880586	3.56403689		
Manga-Andiroba	1.27387648	3.61978175	0.54005340	
Manga-Pracaxi	1.30451783	3.62029692	0.57325153	0.04117519

Murumuru-Açaí	4.74513788	7.19264050	4.63803588	4.42692260
Murumuru-Buriti	3.39740575	7.00088951	3.69479168	3.71034324
Murumuru-Cupuaçu	1.17181258	3.54198816	0.36895693	0.22708130
Murumuru-Jambu	1.11673519	3.57863074	0.34213804	0.28796102
Tucumã-Tucumã	2.82627120	6.19139537	2.91195018	2.83717058
Tucumã-Urucum	0.92625891	3.56553541	0.12767353	0.45104909
Ucuuba-Andiroba	1.60766797	3.43927939	0.86400428	0.59048921
Ucuuba-Buriti	5.40336428	7.15697236	5.71983069	5.77899956
Manga-Pracaxi Murumuru-Açaí Murumuru-Buriti Murumuru-Cupuaçu				
Bacuri-Pracaxi				
Cacau-Tucumã				
Cupuaçu-Açaí				
Cupuaçu-Cacau				
Cupuaçu-Pequi				
Cupuaçu-Tucumã				
Manga-Andiroba				
Manga-Pracaxi				
Murumuru-Açaí	4.41307559			
Murumuru-Buriti	3.71997214	3.77179332		
Murumuru-Cupuaçu	0.25850432	4.55013841	3.77353359	
Murumuru-Jambu	0.32710270	4.56849072	3.72362139	0.12822305
Tucumã-Tucumã	2.83675226	2.63146587	1.50758045	2.92545710
Tucumã-Urucum	0.48753696	4.62252288	3.70385022	0.26820722
Ucuuba-Andiroba	0.56435756	4.49601297	4.06725298	0.62491301
Ucuuba-Buriti	5.77867684	4.92216000	4.72668888	5.82933304
Murumuru-Jambu Tucumã-Tucumã Tucumã-Urucum Ucuuba-Andiroba				
Bacuri-Pracaxi				
Cacau-Tucumã				
Cupuaçu-Açaí				
Cupuaçu-Cacau				
Cupuaçu-Pequi				
Cupuaçu-Tucumã				
Manga-Andiroba				
Manga-Pracaxi				
Murumuru-Açaí				
Murumuru-Buriti				
Murumuru-Cupuaçu				
Murumuru-Jambu				
Tucumã-Tucumã	2.91044572			
Tucumã-Urucum	0.21933882	2.91527405		
Ucuuba-Andiroba	0.75245469	3.07263809	0.82951967	
Ucuuba-Buriti	5.83092235	4.59040079	5.76853456	5.88134722

```
fviz_dist(
  distPair,
  gradient = list(low = "blue",
                  mid = "white",
                  high = "red")
)
```



```
coefHc <- hclust(
  distPair,
  method = "ward.D2"
)
```

coefHc

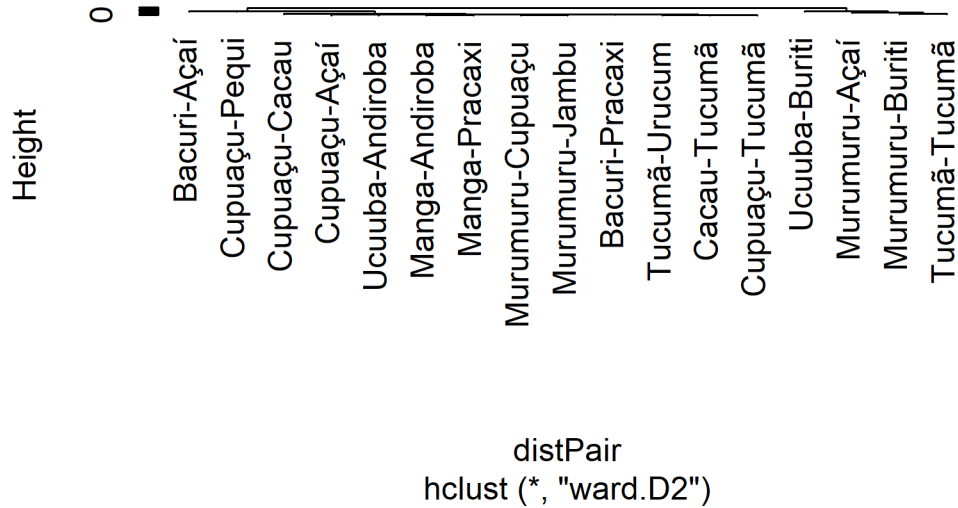
Call:

```
hclust(d = distPair, method = "ward.D2")
```

```
Cluster method : ward.D2
Distance       : euclidean
Number of objects: 17
```

```
plot(coefHc)
```

Cluster Dendrogram



```
pairTree <- cutree(
  coefHc,
  k=3
)
pairTree
```

Bacuri-Açaí	Bacuri-Pracaxi	Cacau-Tucumã	Cupuaçu-Açaí
1	1	1	1
Cupuaçu-Cacau	Cupuaçu-Pequi	Cupuaçu-Tucumã	Manga-Andiroba
1	1	1	1
Manga-Pracaxi	Murumuru-Açaí	Murumuru-Buriti	Murumuru-Cupuaçu
1	2	2	1
Murumuru-Jambu	Tucumã-Tucumã	Tucumã-Urucum	Ucuuba-Andiroba
1	2	1	1
Ucuuba-Buriti			
3			

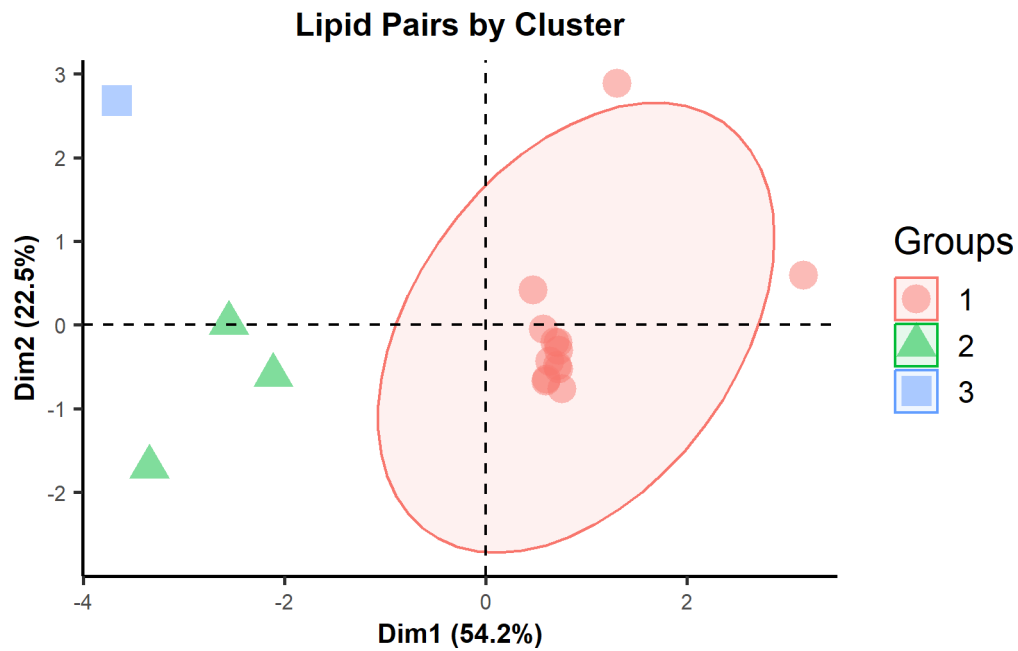
```
pairClut <- fviz_pca_ind(
  X = coefPca,
  geom = "point",
```

```

habillage = factor(pairTree),
addEllipses = TRUE,
mean.point = FALSE,
pointsize = 5,
alpha.ind = 0.5
#title = "Lipid Pairs by cluster"
) + theme_classic(base_size = 14) +
theme(
  plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 12
  ),
  axis.title = element_text(
    face = "bold",
    size = 10
  ),
  axis.text = element_text(size = 8)
) +
labs(
  title = "Lipid Pairs by Cluster"
  #color = "Clusters"
)
pairClut

```

Too few points to calculate an ellipse
Too few points to calculate an ellipse



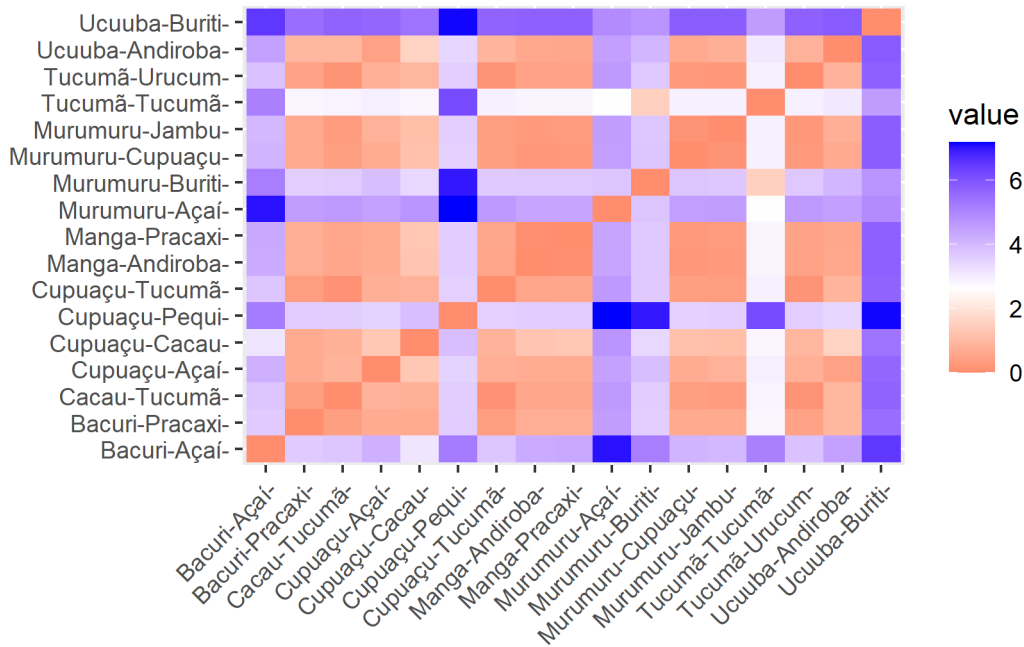
```
ggsave(
  "pairClut.png",
  plot = pairClut,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 600
)
```

Too few points to calculate an ellipse
 Too few points to calculate an ellipse

```
head(coefDataPca)
```

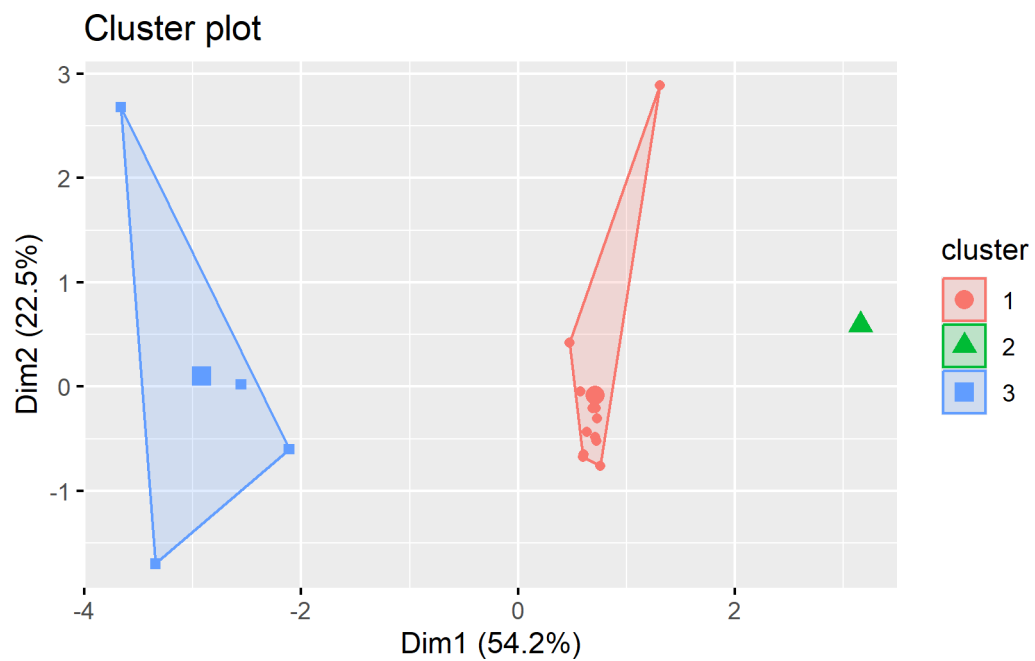
	b1	b2	b3	b12	b13	b23
Bacuri-Açaí	2.3411	-3.464	-1.527	0.00	0.00	0.00
Bacuri-Pracaxi	0.8600	0.410	0.010	0.00	0.00	0.00
Cacau-Tucumã	0.4300	0.270	0.140	0.00	0.00	0.00
Cupuaçu-Açaí	0.7000	1.160	-0.260	0.00	0.00	0.00
Cupuaçu-Cacau	1.2200	-0.280	0.190	0.00	0.00	0.00
Cupuaçu-Pequi	-0.5100	0.830	-1.840	-3.92	7.41	5.15


```
fviz_dist(
  distPair
)
```

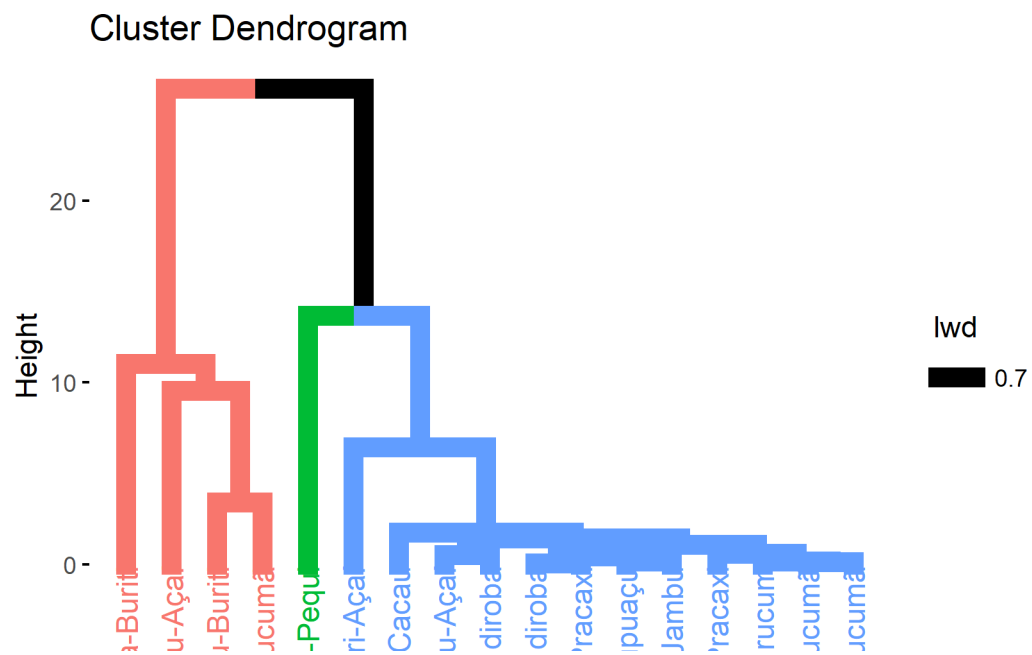


```
coefEhc <- eclust(
  x = coefDataPca,
  FUNcluster = "hclust",
  K = 3,
  k.max = 3
)
```

```
fviz_cluster(
  coefEhc,
  geom = "point",
  repel = TRUE)
```



```
fviz_dend(
  x = coefEhc,
  lwd =
)
```



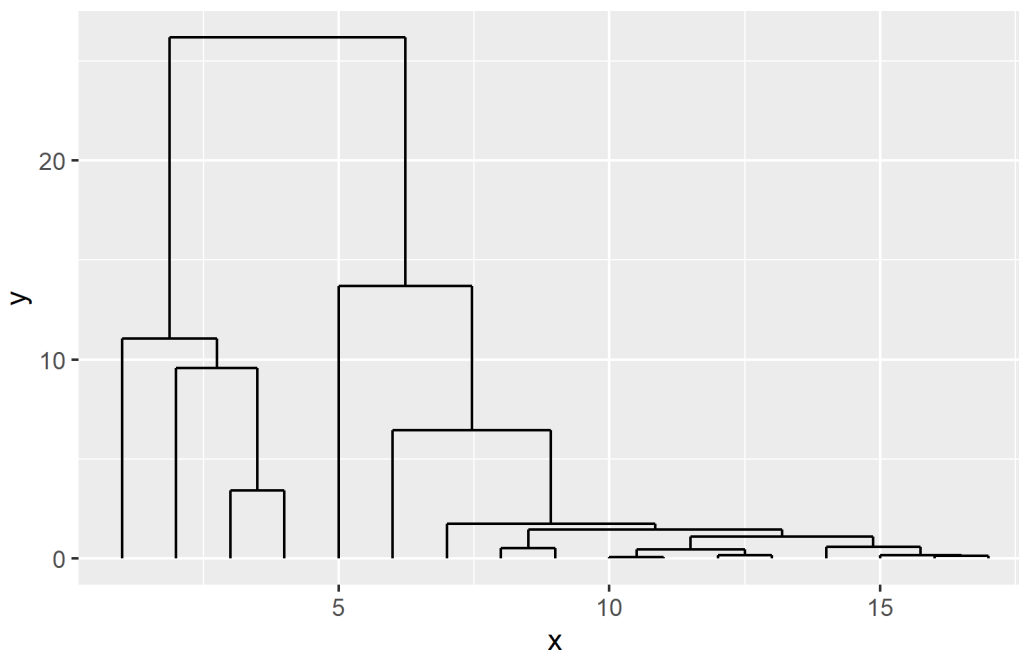
```
library(ggdendro)
```

Warning: pacote 'ggdendro' foi compilado no R versão 4.5.3

```
library(ggplot2)

coefDend <- dendro_data(coefEhc)

ggplot(
  segment(coefDend)
) +
  geom_segment(
    aes(
      x = x,
      y = y,
      xend = xend,
      yend = yend
    )
  ) #+
```



```
#theme_classic()
```

```
#dendPlot
```

```
library(dendextend)
```

Warning: pacote 'dendextend' foi compilado no R versão 4.5.3

Welcome to dendextend version 1.19.1

Type citation('dendextend') for how to cite the package.

Type browseVignettes(package = 'dendextend') for the package vignette.

The github page is: <https://github.com/talgalili/dendextend/>

Suggestions and bug-reports can be submitted at: [https://github.com/talgalili/dendextend/iss](https://github.com/talgalili/dendextend/issues)

You may ask questions at stackoverflow, use the r and dendextend tags:

<https://stackoverflow.com/questions/tagged/dendextend>

To suppress this message use: `suppressPackageStartupMessages(library(dendextend))`

Anexando pacote: 'dendextend'

O seguinte objeto é mascarado por 'package:ggdendro':

`theme_dendro`

O seguinte objeto é mascarado por 'package:stats':

`cutree`

```
coefDend <- as.dendrogram(coefHc)
```

```
coefDend <- color_branches(  
  coefDend,  
  k = 3  
)
```

Carregando namespace exigido: colorspace

```
png(  
  "cluster.png",  
  width = 18,  
  height = 14,  
  units = "cm",  
  res = 600  
)  
  
plot(  
  coefDend,  
  main = "Cluster Dendrogram",  
  xlab = "",  
  sub = "",  
  cex = 1.2  
)  
  
dev.off()
```

```
png  
  2
```